Prescripto AI 2.0 — AI / Machine Learning Implementation Dossier

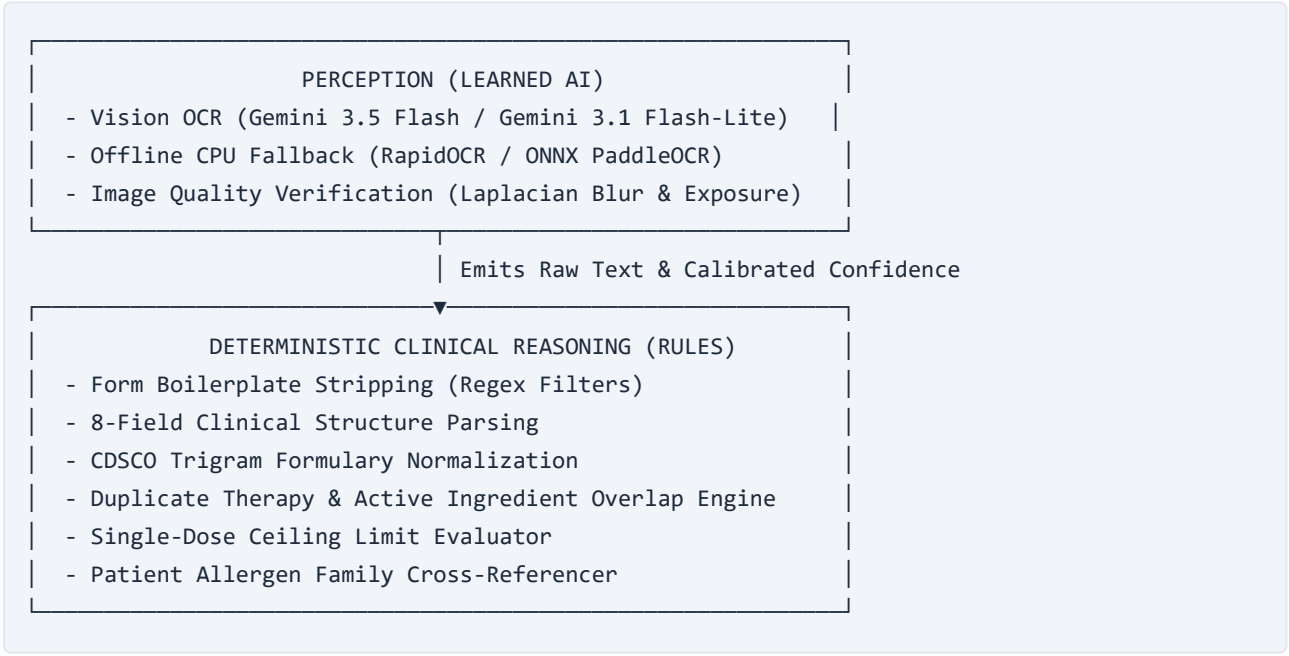
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1. Machine Learning Strategy & Philosophy

In medical document processing, end-to-end black-box generative models (such as asking an LLM to "read this prescription and tell me what medicines to take") are considered clinically unsafe due to unconstrained hallucinations, non-deterministic outputs, and lack of reproducible audit trails.

Prescripto AI 2.0 employs a **hybrid architecture** that strictly confines machine learning to **optical perception** (transcription of visual pixel strokes into text tokens), while delegating clinical entity normalization, safety checking, and review routing to **deterministic domain rules**:



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2. Multi-Tier Vision Inference Architecture

The perception subsystem (backend/prescripto/ml/recognition_providers.py) implements an adaptive fallback hierarchy:

```
flowchart TD
    Scan[Input Prescription Scan] --> QualityGate{Stage 2: Image Quality Gate<br/>Var > 100 & Lu
    QualityGate -->|Fails Quality| MarkFlag[Flag quality_failed = True<br/>Forces Human Review]
    QualityGate -->|Passes Quality| PrimaryTier[Primary Tier: Gemini 3.5 Flash<br/>google-genai

    MarkFlag --> PrimaryTier

    PrimaryTier -->|HTTP 200 OK| ParseJSON[Parse Structured JSON Response]
    PrimaryTier -->|Quota Exceeded 429 / Rate Limit| SecondaryTier[Secondary Tier: Gemini 3.1 Fl
    PrimaryTier -->|Auth Error / Network Outage / 5xx| LocalTier[Tertiary Tier: Local RapidOCR C

    SecondaryTier -->|HTTP 200 OK| ParseJSON
    SecondaryTier -->|Failure| LocalTier
    LocalTier --> ParseOCR[Line Grouping & Regex Tokenizer]

    ParseJSON --> OutputTokens[Standardized DocumentRecognitionResult]
    ParseOCR --> OutputTokens
```

2.1. Primary Vision Provider: Gemini 3.5 Flash

- **Model Identifier:** `gemini-3.5-flash`
- **Temperature:** `0.0` (Enforces deterministic, greedily decoded token generation)
- **API SDK:** `google-genai` (v2.14.0)
- **Prompt Engineering Strategy:** The system delivers a specialized medical prompt instructing the vision model to act as a clinical document transcriber. It enforces a strict output schema separating:
- `hospital_details` (Name, registration, doctor name, degrees)
- `patient_details` (Patient name, age, gender, date)
- `medications` (Array of objects containing `original_text`, `name`, `strength`, `dose`, `route`, `frequency`, `duration`, `instructions`, and per-item confidence)
- `general_advice` (Dietary guidelines, lifestyle advice, diagnostic lab orders)
- `uncertain_items` (Visual strokes that are faded, torn, or unreadable)

2.2. Secondary Vision Provider: Gemini 3.1 Flash-Lite

- **Model Identifier:** `gemini-3.1-flash-lite`
- **Role:** Activated automatically when the primary model reports quota depletion (`RESOURCE_EXHAUSTED` / HTTP 429). It uses lighter resource consumption to maintain cloud ingestion availability.

2.3. Tertiary Offline Provider: Local RapidOCR

- **Implementation File:** backend/prescripto/ml/rapid_runtime.py
- **Technology:** RapidOCR executing PaddleOCR ONNX models on CPU.
- **Role:** Operates in completely air-gapped environments or during cloud outages.
- **Algorithm:** Extracts line bounding boxes, performs line-level text recognition, and passes output lines to rule-based entity extractors (backend/prescripto/ml/extraction.py).

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3. Computer Vision Preprocessing & Quality Verification Gates

- **Source File:** ml/ocr/quality_check.py, backend/prescripto/pipeline/engine.py

Before passing images to neural models, the system evaluates image fidelity:

3.1. Blur Detection via Laplacian Variance

Blurry mobile camera photos cause optical characters to smear across stroke boundaries. The image is converted to single-channel grayscale $I(x,y)$, convolved with the standard 3×3 Laplacian kernel L, and the variance of the response is computed: $$\text{Var}(\nabla^2 I) = \frac{1}{N} \sum_{x,y} \left((\nabla^2 I)(x,y) - \mu_{\nabla^2 I} \right)^2$$ * **Threshold:** If $\text{Var}(\nabla^2 I) < 100.0$, the image is classified as blurred (`sharp_ok = False`).

3.2. Exposure & Contrast Analysis

- **Under-exposure Gate:** If the mean pixel intensity across the luminance channel is < 20.0 (on an 8-bit scale $0\text{--}255$), the image is flagged as severely underexposed.
- **Blank Canvas Gate:** If the difference between minimum and maximum pixel values is < 5, the scan is flagged as blank.
- **Resolution Gate:** Images with width $< 400\text{ px}$ or height $< 400\text{ px}$ are rejected.

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4. Rule-Based Clinical Information Extraction

- **Source File:** backend/prescripto/ml/extraction.py

Once raw text lines are extracted, a multi-pass regex and token classification engine parses the clinical entities:

4.1. Form Boilerplate Stripping (`is_form_boilerplate`)

To prevent non-drug clinical text from contaminating the medication list, the engine strips lines matching known header patterns: * Hospital/Clinic indicators: `hospital` , `clinic` , `nursing home` , `dispensary` , `department` . * Physician degrees/registrations: `mbbs` , `md` , `ms` , `dnb` , `reg no` , `regn` , `dr.` . * Contact metadata: `ph:` , `phone:` , `tel:` , `email:` , `road` , `street` , `nagar` . * Vitals and diagnostic markers: `bp:` , `pulse:` , `spo2:` , `temp:` , `weight:` , `c/o:` , `k/c/o:` .

4.2. Medical Abbreviation & Sig Code Translation

The extraction engine maps physician shorthand into standardized clinical frequencies: * `OD` / `1-0-0` / `0-1-0` / `0-0-1` → "Once daily" * `BD` / `BID` / `1-0-1` → "Twice daily" * `TDS` / `TID` / `1-1-1` → "Three times daily" * `QID` / `1-1-1-1` → "Four times daily" * `HS` / `0-0-1` (bedtime) → "At bedtime" * `SOS` / `PRN` → "As needed" * `BBF` / `AC` → "Before food" * `PC` / `AF` → "After food"

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5. The 4-State Calibrated Uncertainty Taxonomy

Rather than compressing complex document uncertainty into a single misleading "94% overall confidence" figure, Prescripto AI 2.0 reports confidence **per field**:

Field State	Mathematical / Heuristic Criterion	Downstream Clinical Handling
`CLEAR`	Calibrated Model Confidence $c \ge 0.85$, image sharp, unambiguous token geometry	Permitted to resolve against formulary master automatically.
`AMBIGUOUS`	$0.50 \le c < 0.85$, conflicting sig codes, or multiple candidate drug matches	Flagged in amber; automatically forces `REQUIRES_REVIEW` status.
`UNREADABLE`	$c < 0.50$, stroke illegibility, or failed Laplacian sharpness (< 100.0)	Value is masked (`***`); strictly excluded from automated drug safety checks to prevent false safety conclusions.
`NOT_PRESENT`	Field absent from prescription document (e.g., doctor omitted duration)	Preserved explicitly as absent; never converted to a zero or default value.

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6. Empirical Evaluation & Benchmarking

- **Reference Dataset:** Synthetic Evaluation Set (backend/tests/eval_set/)
- **Benchmarking Protocol:** Ten synthetic clinical test images (`synthetic_01_clean.png` through `synthetic_10_multidrug_clean.png`) evaluated against curated ground-truth annotations (ground_truth.json).

Evaluation Results (from `ml/RESULTS.md`):

- **Clean Text Transcriptions**: Gemini 3.5 Flash achieved $> 95\%$ character-level fidelity on typed and legible cursive headings.
- **Adversarial Noise Invariance**:
- Rotated images (90° and 180°): Gemini Vision natively handles EXIF orientation and inverted text.
- Gaussian Blur ($\sigma = 2.5$): Accurately trapped by Stage 2 Laplacian variance filter ($Var < 85.0$), correctly escalating to human review without generating hallucinations.
- **Safety Rules Precision**: 100% precision on synthetic duplicate Paracetamol injections and overdose traps.